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PV-LAB

(LABORATORY OF PHOTOVOLTAICS AND THIN-FILMS ELECTRONICS)

Report of internship:

Assessing the potential of solar-powered EV
charging stations in Dakar

Author:

Talla DIAGNE

Supervisors:

Jérémy DUMOULIN

Nicolas WYRSCH

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Introduction

Dakar, the capital of Senegal in West Africa, faces severe mobility challenges. Rapid demographic growth and urban concentration have led to chronic traffic congestion, with over 7.2 million trips made by the population over 11 years old each weekday [1]. These conditions contribute to high pollution levels and inefficiencies in daily commuting, demanding urgent sustainable solutions. Dakar is the leading region in Senegal in terms of demographics, economics and infrastructure. It is home to the majority of the country's population and activities. According to the Regional Service of Statistics and Demography (ANSD in French), in 2022, Senegal's Gross Domestic Product was valued at 17,227.9 billion FCFA. An analysis of its distribution across regions reveals significant disparities. Indeed, the Dakar region concentrates 46.2% of the national wealth with a GDP of 7,960.8 billion FCFA. This situation is explained by the high concentration of economic units in the region [2, page 3]. Administratively, it is composed of four departments: **Dakar**, **Pikine**, **Guédiawaye**, and **Rufisque**.

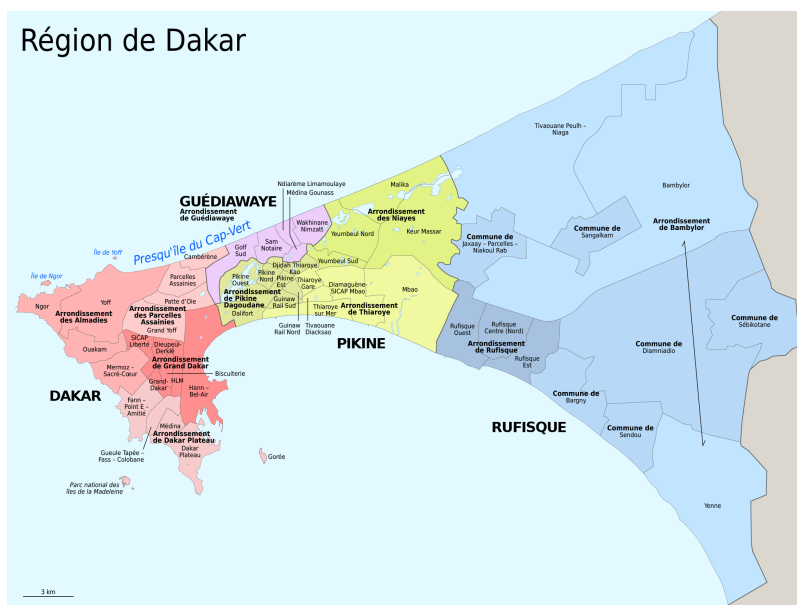


Figure 1: Map of Dakar

This urban concentration places considerable pressure on the city's mobility infrastructure. To address these challenges, public authorities have prioritized strengthening collective transport systems. The Urban Mobility Plan (PMUD) emphasizes high-capacity modes such as the electric Bus Rapid Transit (BRT) and the Regional Express Train (TER) [3, 4]. Currently, public transport is widely used, with minibuses ("Tatas"), rapid buses, and clandestine taxis ("clandos") serving as the main modes [1]. Despite this shift toward collective transport, private cars still account for the majority of vehicles on the road, representing nearly two-thirds of traffic volume [5]. Household motorization increased from 169 to 223 vehicles per 1,000 households between 2015 and 2019 [5]. As this private fleet continues to grow, discussions are increasingly turning toward its electrification as a way to reduce emissions and reliance on fossil fuels. Senegal is implementing a national strategy

to support this transition, aiming to reach 37,000 EVs in circulation by 2030, with a penetration rate of 15%. Measures include exemptions from import duties, reducing vehicle purchase costs by 27%, removal of taxes on spare parts, and a major investment in infrastructure, with 3,700 charging stations planned and an annual budget of 9 billion CFA francs [6].

However, the move toward electric vehicles (EVs) brings important challenges that Dakar must prepare for. One of the biggest issues is charging: there are currently only five public stations in the city [7], and setting up private ones is often expensive and not practical for many households. The power grid will also face pressure, as charging many vehicles at the same time could overload the system unless it is upgraded. On the environmental side, electric cars can reduce air pollution, but their real benefit depends also on how the electricity is produced. Finally, the high cost of vehicles and the lack of equal access to infrastructure risk leaving parts of the population behind, making public support and clear policies essential.

Addressing these challenges requires innovative solutions that can overcome limited charging infrastructure, high costs, and grid constraints. One promising approach is the deployment of shared solar-powered charging stations. This study leverages the EVPV-Simulator[8], an open-source modeling framework that integrates mobility simulation, EV charging demand, and photovoltaic (PV) generation to evaluate synergies between electric mobility and solar energy. In addition to the standard EVPV approach, we analyze urban mobility demand and explore different charging scenarios, while introducing a new smart charging strategy designed to maximize photovoltaic self-consumption and reduce peak loads by dynamically shifting charging loads to periods of high solar production. This approach allows for an optimized assessment of how shared solar-powered EV charging stations could be deployed in Dakar, balancing the need to reduce charging costs, minimize carbon emissions, and limit peak load on the grid.

The report is structured to first present the methodology and the EVPV-Simulator modeling framework. This is followed by an analysis of the simulation results, including mobility patterns, charging demand, and smart charging strategies. It then concludes with a discussion of the benefits and recommendations for deploying solar-powered charging stations in Dakar.

1 Methodology

This section presents the modeling approach used to evaluate the integration of EVs and solar PV in Dakar. The methodology is based on the EVPV-Simulator, which combines mobility simulation, charging demand estimation, and PV production modeling to assess the potential synergies between EV charging and solar energy.

1.1 EVPV-Simulator

The EVPV-Simulator[8] is an open-source tool developed to evaluate the charging needs of EVs and the potential contribution of solar PV systems to meet these needs. The model estimates mobility demand by simulating trips within defined traffic zones based on population density, workplaces, and points of interest. It then calculates the spatial and temporal distribution of EV charging requirements and assesses how much of this demand could be supplied by local solar generation. The modeling framework is therefore organized into three successive stages, each of which is detailed in the subsections of [1.3]. The first stage, *Mobility demand*, quantifies daily mobility flows and travel distances using a spatially approach. The second stage, *Charging demand*, translates these flows into spatio-temporal charging requirements by integrating vehicle characteristics, infrastructure

properties, and user charging behaviors. The third stage, *EV-PV complementarity*, evaluates the extent to which local photovoltaic production can contribute to satisfying the identified charging needs [9].

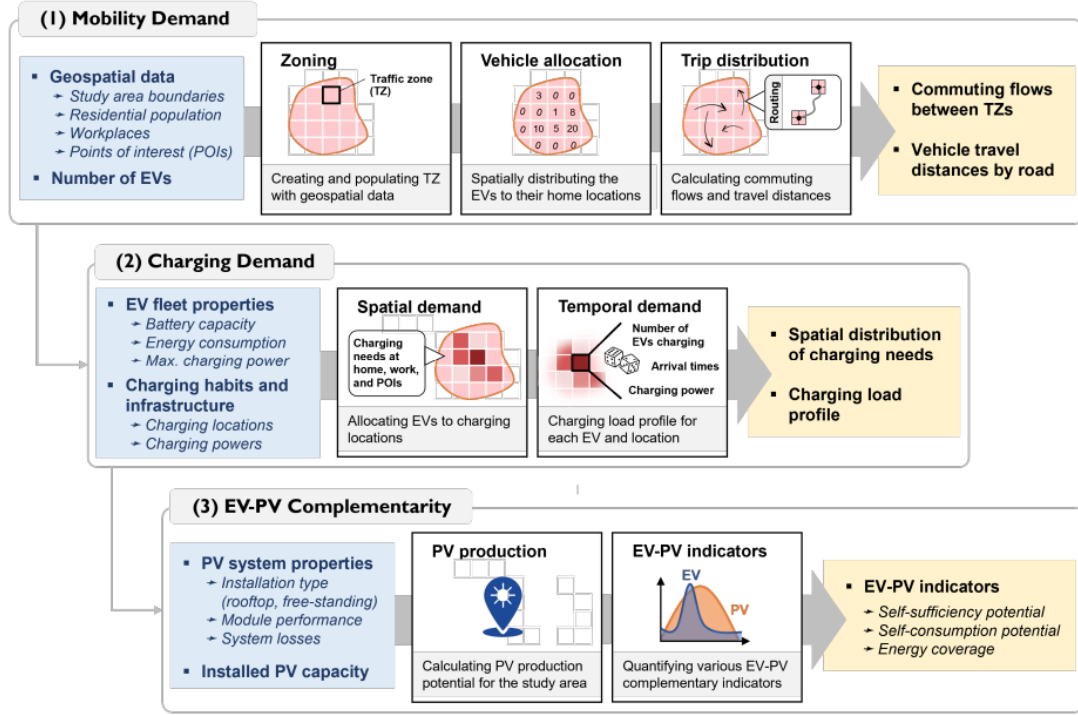


Figure 2: Overview of the three main steps of the modeling framework, [9, page 5]

Adapting the EVPV-Simulator to Dakar involves collecting local geospatial data, calibrating the model to reflect the characteristics of the vehicle fleet and charging habits, and developing scenarios to simulate different levels of EV adoption and PV integration. The results provide insights into optimal strategies for combining electric mobility with solar energy in Dakar’s urban context.

1.2 Study area

The study focuses on the Dakar metropolitan region (Figure [1]), including the city center and its surrounding urban areas. For the EVPV-Simulator analysis, the region is divided into a spatial grid of traffic zones, each approximately 1 km^2 . This resolution is designed to capture the main features of urban dynamics, including population distribution, workplaces, and points of interest. This spatial framework serves as the basis for simulating mobility patterns, vehicle allocation, and charging infrastructure needs in Dakar.

1.3 Input data & parameters

The EVPV-Simulator requires several types of input data. The modeling framework assumes a mixed fleet of battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs), daily commuting flows generated with a gravity model, and solar generation estimated from local irradiance and PV system parameters.

1.3.1 Mobility demand

The first step quantifies daily electric vehicle mobility using a spatially detailed model. The study area is divided into traffic zones, with a specific number of EVs assigned to each zone. A trip distribution model then simulates flows and travel distances between zones. The model focuses on weekday mobility, emphasizing home-to-work trips and their charging needs at home, work, and public stations [9].

The simulation uses population rasters (GHSL/POP [10], Figure[3]) and administrative boundaries (GADM [11]) to define traffic zones and estimate the number of residents in each zone. In the base model, vehicles are initially allocated uniformly across zones. A gravity-based model distributes trips between zones based on the attractiveness of workplaces and points of interest, as well as road distance costs. To better reflect the strong geographic imbalance in Dakar, a spatial allocation of vehicles is applied as detailed in the Appendix[C]. Recent ANSD data [12] confirm this disparity: in 2019, the Dakar department accounted for 76.2% of vehicles, followed by Pikine (13%), Guédiawaye (6%), and Rufisque (4%), even though the most densely populated departments are Guédiawaye and Pikine, as shown in Figure[3]. For this study, focusing on the 2035 horizon, a target of 40,000 EVs is considered for the Dakar region. This setup generates detailed spatial and temporal mobility flows, capturing peak commuting periods, which are then used to determine when and where EVs will require charging throughout the metropolitan area.

When calculating travel distances between traffic zones, the algorithm returns the euclidian distance. It is then primordial to get, from this value, the actual travel distance by road. First, we can verify if the euclidian distance seems correct in comparison with litterature. In their article, Louf et al. [13] found an average commuting distance of 4.6 km for the region of Dakar. Since this value corresponds to the distance between cell towers, it is sufficient to set the road to euclidian ratio to 1 in the EV-PV model for comparison, which yields an average commuting 4.04 km. This suggests that the distribution model performs reasonably well. According to the study by Dong and Ruiqi Li [14], the ratio between road distance and euclidean distance (calculated using the Google Maps API on 100,000 points) in Dakar is 1.46, meaning that actual trips are on average 46% longer than the straight-line distance obtained in the previous paragraph. Using the *compare_road_and_euclidian_distance* module of EVPV, which calculates the ratio on 20 points using the Open Route Service API, I obtained 1.48. This value can therefore be validated and used for subsequent simulations.

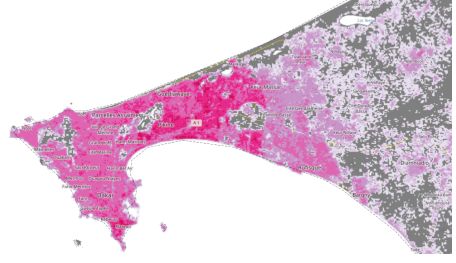


Figure 3: Population raster file for Dakar, [10]

1.3.2 Charging demand

The next step in the modeling framework focuses on the spatio-temporal distribution of electric vehicle charging needs and the estimation of the number of charging points required. The approach considers the properties of each vehicle type, including battery capacity, energy consumption per kilometer, and maximum charging power. It also accounts for user charging habits, such as the proportion of EVs charging at home, at work, or at public stations, as well as the characteristics of the available charging infrastructure, including the power that can be delivered at each location.

The simulated fleet includes both BEVs and PHEVs. BEVs are modeled with a battery capacity of 60 kWh, a consumption of 0.18 kWh/km, and a maximum charging power of 100 kW, while PHEVs have a 15 kWh battery, the same consumption rate, and a maximum charging power of 11 kW [9]. Charging behavior is assumed to be possible either at work, at home or at points of interest, and a range of charging power options from 7.4 kW up to 50 kW. Arrival times at charging locations are modeled using probability distributions that reflect typical commuting patterns, with most vehicles arriving at work around 9:00 AM (± 1 hour) and returning home around 6:00 PM (± 2 hours).

1.3.3 PV potential

Before quantifying Senegal’s solar resources, it is useful to highlight the relevance of PV production for electric mobility. In particular, the EVPV model considers the complementarity between local PV generation and EV charging, by estimating how far solar energy can directly cover mobility-related electricity demand.

According to [15, page 13], solar resources in Senegal are characterized by about 3,000 hours of sunshine per year and an average daily solar radiation of 5.8 kWh/m²/day. So far, these resources have been used through photovoltaic and solar thermal systems, but Senegal still exhibits a significant reliance on fossil fuels (70 %), while photovoltaic accounts for only 11 % of the electricity mix in 2022 [16, page 21]. However, since their first installation in the 1980s [15], PV solar systems have seen growth in Senegal, being used in various areas like telecommunications, water pumping and both centralized and decentralized electricity production. By 2000, the total installed capacity was 2 GWh. In 2020, Senegal produced 385 GWh of solar energy, accounting for 6.63% of its total energy production. Still according to [16], solar PV production reached 582 GWh in 2023. These trends highlight the strong potential of solar PV in order to support the electrification of transport in Senegal.

Using the specific yield provided by the EVPV model ($SY = 1,789 \text{ kWh/kWp/year}$), the average daily production per unit of installed power is $p_{\text{day}} = 1789/365 \approx 4.9 \text{ kWh/kWp/day}$. For 40,000 EVs, each traveling 10 km/day with a consumption of 0.18 kWh/km, the daily demand is $D_{\text{day}} \approx 80 \text{ MWh/day}$, with a maximum of 83 MWh/day (see Section[2.3]).

The required nominal PV capacity is therefore :

$$P_{\text{req}} = \frac{D_{\text{day}}}{p_{\text{day}}} \approx 17 \text{ MWp},$$

and we therefore choose 17 MWp as the PV capacity for the simulations, which is negligible compared to the total national installed capacity of 226 MW according to [17].

1.4 Charging scenarios

To explore the impact of charging behavior on grid and PV integration, four main charging scenarios are considered. These scenarios combine two key factors: the location of charging -either at home or at the workplace- and the charging strategy -either "dumb" (uncontrolled) or smart (optimized to reduce peaks while improving self-consumption). Table [1] summarizes the four combinations. Scenario S1 represents home charging with a dumb strategy, while S2 applies a smart charging strategy at home. Scenario S3 considers dumb charging at the workplace, and S4 combines workplace charging with smart management. These scenarios provide a framework to analyze how both location and charging management affect spatial and temporal electricity demand in Dakar.

- Charging location: home vs. workplace
- Charging strategy: "dumb" vs. smart charging

Scenario	Charging location	Charging strategy
S1	Home	Dumb
S2	Home	Smart
S3	Workplace	Dumb
S4	Workplace	Smart

Table 1: Combinations of charging scenarios considered

2 Results and discussion

2.1 Mobility patterns

Figure 4 shows the distribution of average daily commuting distances. The simulation assumed a 100 % workplace charging scenario. Under this assumption, the average round-trip commuting distance was 5.05 km. This value is consistent with the model resolution: considering that distances are round trips, the mesh grid used here is 1 km and the average road-to-euclidean distance ratio is around 1.48, it is expected that no trips are shorter than 3 km. The longest travel distance measured is 25 km, which means a one way euclidean distance of $\simeq 8.5$ km. This is relatively short, considering that the Dakar region stretches over roughly 30 km from east to west. Nevertheless, it is consistent with the general tendency of people to live close to their workplace.

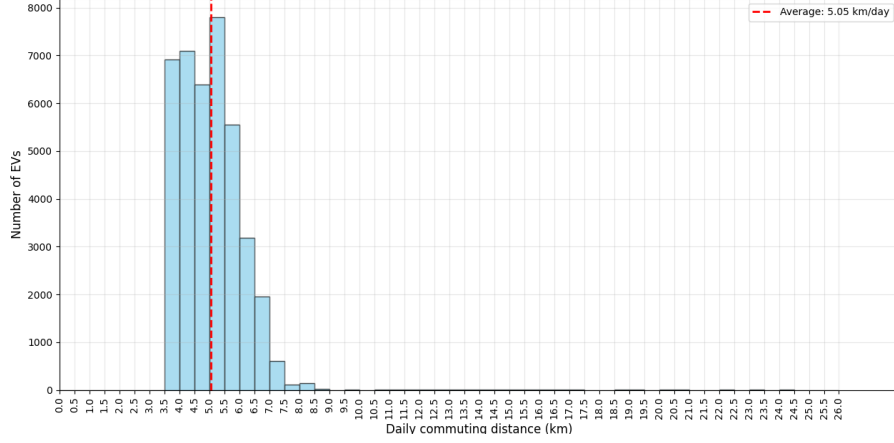


Figure 4: Histogram of daily round-trip commuting distances (km)

The mobility pattern is also analyzed by mapping the inflows and outflows of vehicles for each traffic zone. Figure[5a] and Figure[5b] respectively show the outgoing and incoming flows, still under a mixed charging scenario. The mapping of outflow by area [5a] shows a predominance of the departement of Dakar in the Westside of the region. This result is a direct consequence of the vehicle allocation discussed in the Appendix [C], combined with the population density. As previously shown in Figure [3], the Plateau area has a particularly high population density.

As for the inflow map [5b], while some vehicle flows are directed towards the eastern departments such as Rufisque, the vast majority of trips remain concentrated within the Dakar department. Notable hotspots of incoming flows are observed in Grand Yoff and Point E. The prominence of Point E can be explained by several complementary factors: the relocation of companies from the congested Plateau, its high centrality attracting entrepreneurs due to favorable environment, socio-economic status of residents, and accessibility, and the development of commercial and leisure activities such as restaurants, art galleries, and sports clubs [18, page 88]. This suggests that the simulation captures realistic mobility patterns. The highest inflow value to the east corresponds to SEDIMA, an industrial site specializing in poultry farming, which is also identified multiple times in OpenStreetMap (OSM). Compared to Figure[15] discussed in appendix, we can see that, globally, the west is still way more attractive the eastern part of the region. However, the simulation predicts a higher outflow than inflow for Plateau, for example. This could be interpreted as a repulsive area, even though Plateau is shown in Figure 15 as the most attractive zone in the region. This is due to the allocation of vehicles. Another map without the allocation can be found on Appendix[B].

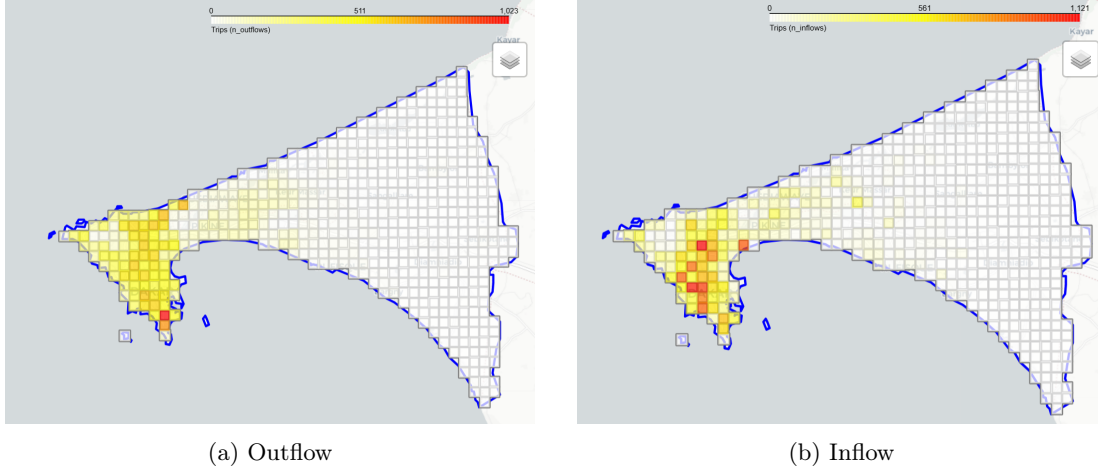


Figure 5: Mapping of incoming/outgoing flows by zone. Results are similar for both home/workplace charging scenarios

2.2 Total EV charging needs and spatial distribution

Figure 9 shows the daily balance between PV generation peak and EV charging demand peak over the course of one year under a mixed charging scenario. Blue points correspond to days when PV production exceeds EV demand, indicating potential surplus energy that could be stored or exported. Red points correspond to days when EV demand surpasses PV generation, requiring electricity from the grid or from storage. The yearly profile clearly reflects the seasonal pattern in Senegal: during the dry season (roughly November to May), PV production is higher and daily surpluses are more frequent. During the rainy season (roughly June to October), reduced solar irradiance results in more and more unmet charging demand by the PV production. Actually, as shown in Figure 6, the period begins slightly earlier, around May. Overall, the yearly demand is met on only about 45% of the days. Among the days when the demand is not met, 86% of the deficits are less than 2 MW. This is a direct consequence of the specific yield value used in Section[1.3.3] and which was returned to me by the EVPV model : it's an annual value that I used to have the average daily production per unit of installed power $p_{day} = 4.9 \text{ kWh/kWp/day}$. However, according to Niang et al.[19], the actual performance of solar plants in Senegal shows strong seasonal variability, with daily yields ranging from about 3.9 kWh/kWp/day during the rainy season to over 7.5 kWh/kWp/day in the dry season. This highlights that using a single annual average value may underestimate the contribution of PV during peak production periods and overestimate it during low yield months.

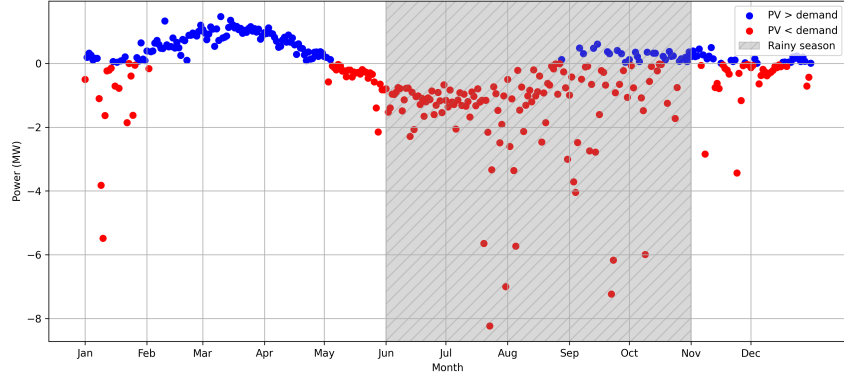


Figure 6: Daily PV generation versus EV charging demand over one year: workplace charging scenario

Figure[7] illustrates the spatial distribution of charging demand for two extreme scenarios: all vehicles charging at home (Scenario S1, Figure[7a]) and all vehicles charging at the workplace (Scenario S3, Figure[7b]). In both cases, the demand is way higher in the western part of the region, as expected when allocating non-evenly the cars. When charging at workplaces, electricity demand is heavily concentrated in the city center- the main employment hub - where it peaks at 2.3 MWh per zone. By contrast, under a home-only charging scenario, demand is far more dispersed across the Dakar department, with peaks located around Dakar-Plateau (see Figure [1]) and a maximum value at 1.9 MWh per zone. The disperse peaks observed in the workplace could be expected because we use OSM tags as attraction centers, whereas in the home scenario, the population raster is less clustered. In this case, the home scenario seems to be a better option if we want to avoid grid stress. Even though Guediawaye and Pikine are very densely populated, we can see on these maps that the demand low. This is mainly due to the fact that both departments combined share only 18% of the personal vehicles in the region. For the workplace scenario, we can observe similarities in the hotspots already seen in the inflow map on Figure[5b].

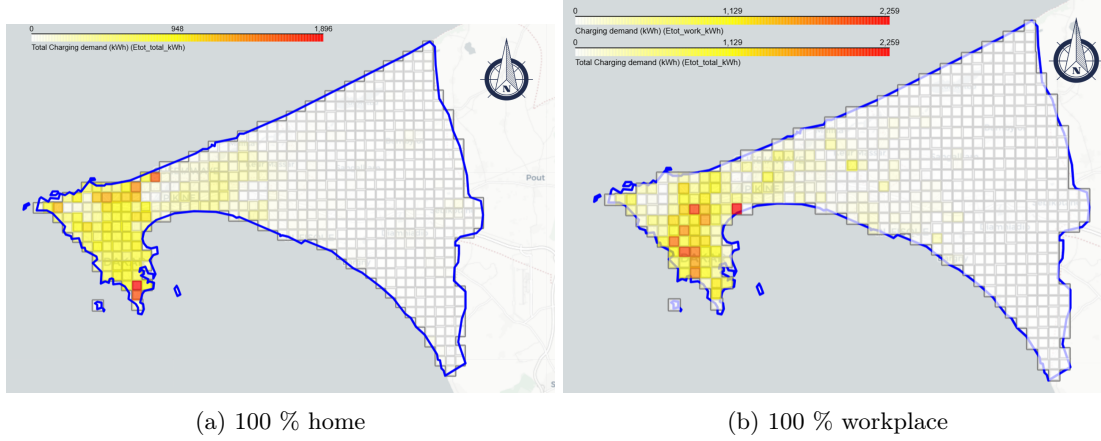


Figure 7: Mapping of charging demand by scenario in Dakar

2.3 Charging load under different charging strategies

Smart-charging is implemented in this study through a custom code, referred to as **maxSC**. The objective of this algorithm is to maximize photovoltaic self-consumption by dynamically shifting charging demand toward periods of high PV generation. Unlike uncontrolled ("dumb") charging, used so far in this report and which starts as soon as vehicles arrive at their charging location, the smart-charging strategy works differently. It delays or distributes charging sessions within the available parking window to better align with solar production. At the same time, it ensures that each vehicle still reaches its required state of charge before departure. If that goal is not achieved, the rest of the charging demand is evenly distributed over the entire time slot of the car's presence.

The temporal profiles of EV charging demand and PV production are analyzed for different charging strategies and locations. Figures[8] and [9] show load curves for scenarios where all vehicles charge at the workplace (Scenarios S3 and S4) or at home (Scenarios S1 and S2), respectively. With the uncontrolled charging strategy, peak demands coincide with commuting hours, with workplace charging producing sharper morning peaks and home charging producing more evenly distributed evening peaks.

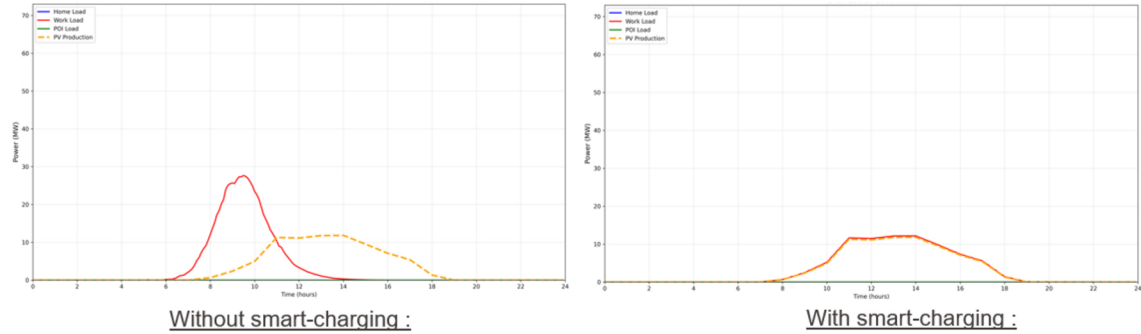


Figure 8: PV production and load demand curves: 100% workplace charging scenario

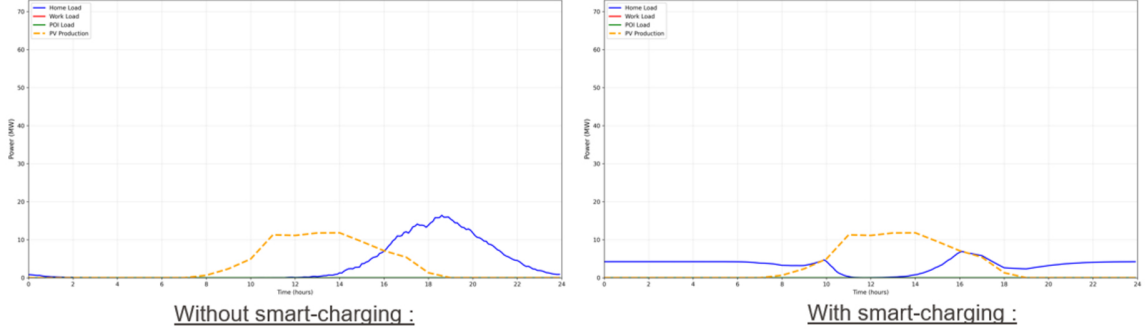


Figure 9: PV production and load demand curves: 100% home charging scenario

Figures [10] illustrate how charging demand interacts with PV production on different days. The load curve on January 1st is compared with that of August 31st. August corresponds to the rainy season in Senegal, when PV production is usually lower and more variable than in January. By comparing these two days, we can evaluate how effectively the smart-charging strategy shapes the charging demand to follow the PV production curve.

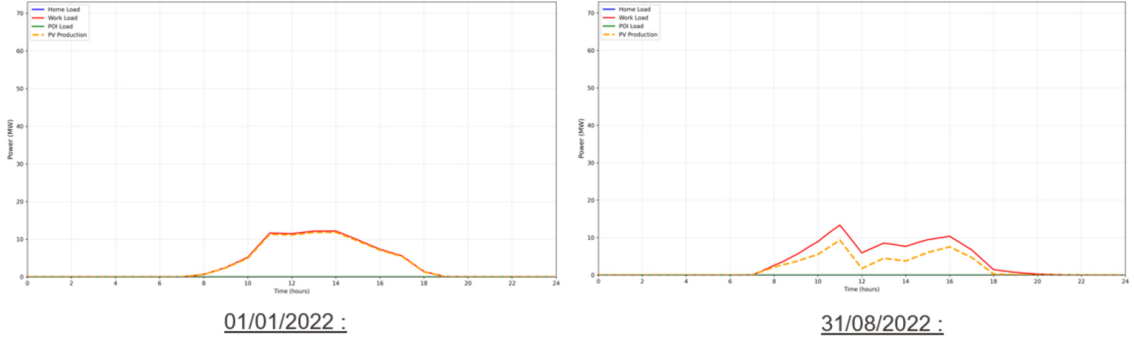


Figure 10: PV production and load demand curves: 100% workplace for different days

Figure[11] compares the self-consumption and self-sufficiency achieved under different charging scenarios and strategies, using an annual simulation. The charging demand peak value was retrieved for the January 1st, for a PV nominal capacity of 17 MW. Self-consumption represents the fraction of locally generated PV electricity directly used to charge EVs, while self-sufficiency indicates the fraction of the total EV charging demand covered by local PV generation. The formulas used are recalled in Appendix[D]. The figure below shows that workplace charging combined with the smart-charging strategy significantly improves both self-consumption and self-sufficiency compared to home charging or uncontrolled ("dumb") charging. Unlike workplace charging, where the smart strategy can double the self-consumption value, home charging shows almost no improvement, which makes sense because at home EVs are usually charged later in the day when there is less sunlight.

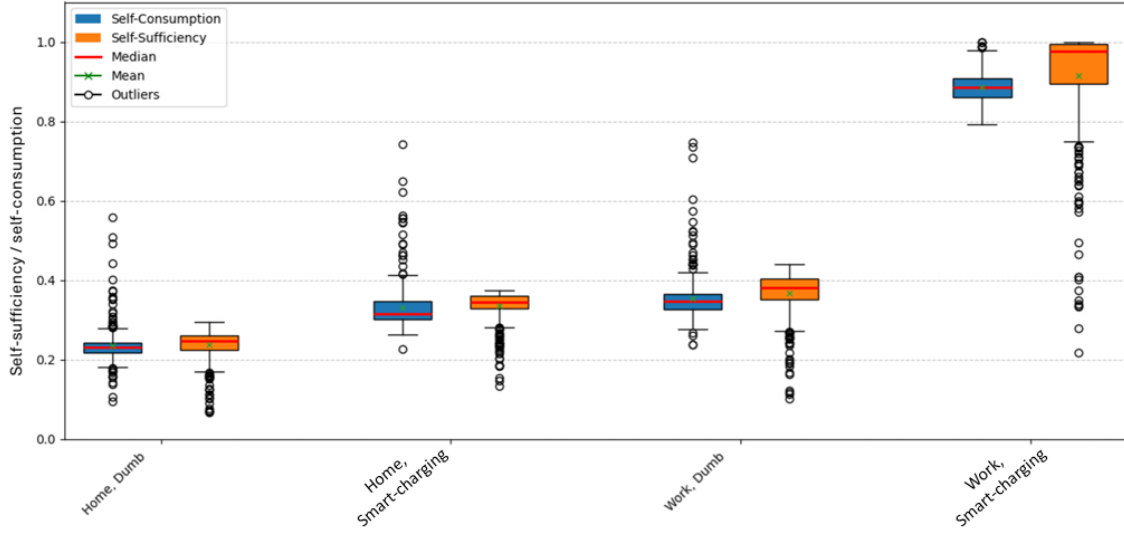


Figure 11: Comparison of self-consumption and self-sufficiency across charging scenarios and strategies; annual simulation with peak values on 01/01, 0.425 kW/EV

2.4 Environmental and economic benefits

The transition from internal combustion engine vehicles (ICEVs) to EVs offers both environmental and financial advantages. The following sections detail the estimated CO₂ emission reductions and cost savings per vehicle based on the simulated travel patterns from the EVPV model.

2.4.1 CO₂ emission reductions

Electric vehicles can significantly reduce greenhouse gas emissions compared to ICEVs. To quantify this benefit, ICEV emissions were calculated using standard per-kilometer values, while EV emissions were estimated based on electricity consumption per kilometer, charging efficiency, and the carbon intensity of Senegal's electricity mix. According to the United for Efficiency country assessment, the national grid has an emission factor of about 0.535 kg CO₂/kWh in 2023 [20].

The results for a 100% home charging scenario are shown in Figure 12. We then obtain a total emission of 8,288,844.39 kg CO₂ avoided for the entire fleet, or 207.22 kg CO₂ per vehicle per year. For a 100% workplace charging scenario, the results are very similar. When considering the use of local PV to cover the EV charging demand at the workplace, we estimate the PV carbon intensity (CI) by adapting a reference measured in Switzerland (42.5 gCO₂/kWh [21]) to Dakar using the ratio of expected energy yields. This yields a CI of approximately 23 gCO₂/kWh for Dakar. With this PV electricity, the CO₂ savings per vehicle increase to 590 kg CO₂ per year, nearly tripling the benefit compared to charging from the grid.

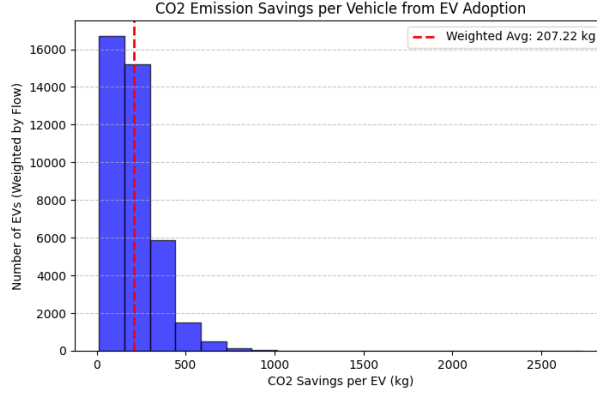


Figure 12: Annual CO₂ emission savings per vehicle from EV adoption, 100% home charging scenario

2.4.2 Economic benefits

In addition to environmental advantages, EVs can provide significant cost savings for users. ICEV fuel costs were calculated using the national average gasoline price of 1.769 USD per liter [22], while EV operating costs were based on the residential electricity price of 0.18 USD per kWh [23]. These values allow us to estimate the financial savings that EV users can achieve compared to ICEVs.

For a 100% home charging scenario, the average annual savings per vehicle are shown in Figure [13]. Each EV saves \$386.74 per year, which is more than twice the average monthly urban salary in Senegal (104,869 FCFA $\simeq$ \$187.10) according to [24], using the exchange rate at the time (\$1 $\simeq$ 560.79 FCFA). These calculations are based on residential electricity rates. Given this price, there is therefore room for companies wishing to install charging stations to make a profit. Moreover, the tariff from Senelec (the electricity company in Senegal) depends on the consumption tier, so using an EV should also take into account the potential move to a higher tier, which could increase costs. For example, by using post-payment Woyofal, which covers approximately 71% of the population [25], the cost for a Medium Power Domestic (Domestique Moyenne Puissance) connection could increase by around 30% [26].

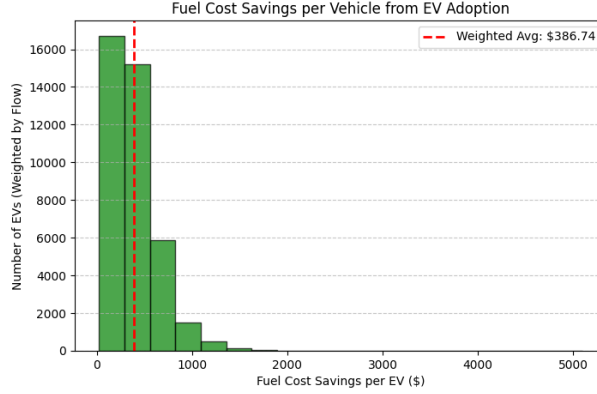


Figure 13: Annual fuel cost savings per vehicle from EV adoption, 100% home charging scenario

Taken across the entire simulated fleet, these savings represent a significant opportunity for Dakar to lower transport costs while simultaneously reducing its carbon footprint.

3 Conclusion and recommendations

This study assessed the integration of EVs with PV in Dakar using the EVPV-Simulator, introducing a new smart charging strategy to maximize self-consumption. By combining mobility modeling, charging demand, and PV generation, the analysis highlights both the opportunities and the data challenges of electrifying Dakar’s fleet.

The analysis reveals several important insights regarding the integration of EVs and solar energy in Dakar:

- The smart-charging strategy in the workplace charging scenario most effectively aligns demand with PV generation, with a yearly mean self-consumption of around 88.5%, compared to 35.4% for the dumb scenario
- Charging demand is concentrated in central Dakar peaking at 2.3 MWh for the workplace charging scenario, highlighting the need for targeted infrastructure planning
- EV adoption offers significant economic benefits, with annual savings per vehicle exceeding twice the average monthly urban salary (\$386.74/EV/year, [24])
- Using EVs reduces emissions by 207.22 kg CO₂/year per vehicle from the grid; using local PV for workplace charging increases savings to 588.60 kg CO₂/year

A decentralized approach –installing PV systems and EV charging infrastructure directly at workplaces for example– can locally match electricity supply with demand, reduce transmission losses, and better integrate variable solar generation. This strategy maximizes both economic and CO₂ savings per vehicle, while smart charging and supportive policies further enhance these benefits. Targeted planning in central Dakar remains essential to ensure infrastructure aligns with concentrated charging demand.

At the same time, this study faces several limitations. The analysis relies on modeled mobility flows and average assumptions for driving and charging behaviors, which may not fully reflect actual practices. The uneven distribution of vehicles across departments and the limited availability of detailed mobility data constrain the accuracy of the results, while the use of OpenStreetMap data for workplace locations and points of interest introduces further uncertainty.

Future work should therefore focus on improving local datasets, exploring the optimal spatial distribution of charging stations, assessing the impact of dynamic electricity tariffs and grid constraints, and extending the framework to other regions of Senegal.

Appendices

A Tags used for POIs and workplaces

```
1  'workplaces': {
2      'tags': {
3          "company": True,
4          "industrial": True,
5          "office": True,
6          "landuse": ["industrial", "commercial", "retail", "education", "
institutional"],
7          "building": ["industrial", "commercial"],
8          "tourism": ["hotel", "hostel", "motel", "guest_house"],
9          "amenity": [
10             "university", "college", "school", "research_institute",
11             "hospital", "clinic", "doctors", "dentist",
12             "pharmacy", "laboratory", "townhall", "courthouse",
13             "embassy", "police", "fire_station", "post_office",
14             "prison", "bank", "money_transfer",
15             "marketplace", "internet_cafe", "coworking_space",
16             "conference_centre", "library", "training",
17             "townhall", "restaurant", "bar", "cafe", "fast_food", "food_court",
"ice_cream"
18         ],
19     },
20     "shop": [
21         "supermarket", "mall", "department_store", "wholesale",
22         "bakery", "butcher", "electronics", "hardware",
23         "furniture", "computer", "books", "stationery",
24         "laundry", "dry_cleaning", "hairstylist", "beauty",
25         "tailor", "florist", "optician", "jewelry",
26         "gift", "sports", "music", "photo",
27         "tobacco", "ticket", "copyshop", "pet"
28     ],
29 },
30 },
31 }
32
33 "POIs": {
34     "tags": {
35         "amenity": [
36             "fuel", "parking", "parking_entrance", "bicycle_parking",
37             "college", "university", "school", "kindergarten",
38             "library", "music_school", "language_school",
39             "clinic", "dentist", "doctors", "hospital", "pharmacy",
40             "veterinary", "cafe", "ice_cream", "internet_cafe",
41             "restaurant", "fast_food", "bar", "pub", "biergarten",
42             "theatre", "cinema", "music_venue", "nightclub",
43             "casino", "gambling", "stripclub", "arts_centre",
44             "community_centre", "social_centre", "exhibition_centre",
45             "attraction", "viewpoint", "aquarium", "beach_resort",
46             "gallery", "museum", "theme_park", "zoo", "artwork"
47         ],
48         "shop": [
49             "supermarket", "mall", "department_store", "convenience"
50         ],
51     },
52 }
```

```

51     "tourism": [
52         "hotel", "guest_house", "hostel", "motel",
53         "camp_site", "apartment"
54     ],
55     "leisure": [
56         "stadium", "sports_centre", "swimming_pool", "fitness_centre"
57     ]
58 }
59 }

```

The economic structure of the Dakar region in 2023, as documented by ANSD [27], shows a strong dominance of the tertiary sector. Services form the main pillar of the regional economy, accounting for 51.7% of activities, followed by trade and repair (22%) and industry (24.4%). The agricultural sector represents only a marginal share of 1.9%, reflecting the urban and service-oriented nature of Senegal's capital.

The tags, recalled justa above, were classified according to their category and assigned a weight to match the official statistics. Given its very low share, the agricultural sector was not considered. After the first run of the simulation of total inflows per zone using EVPV, the results are shown in the image on the left:

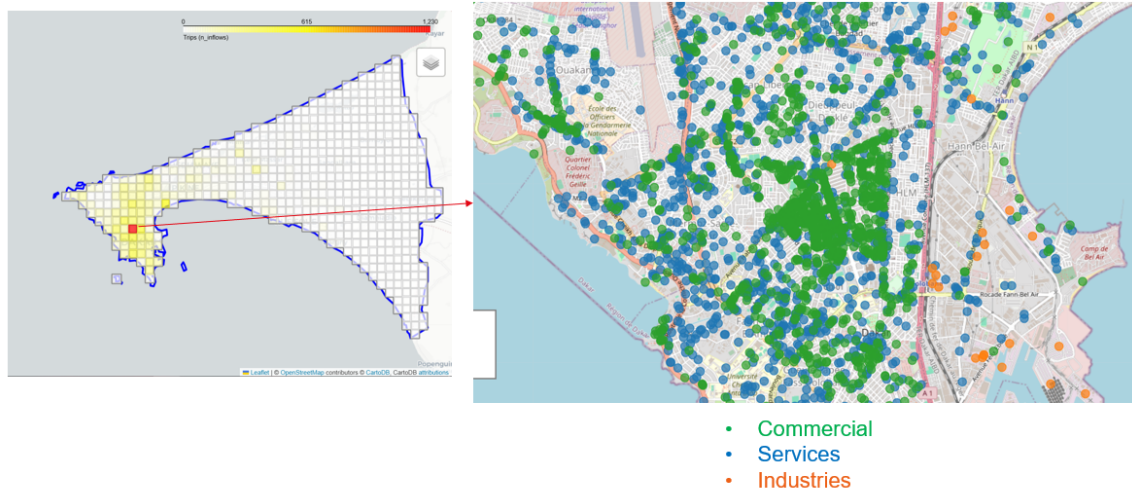


Figure 14: Simulation of the number of inflows per zone and maps of workplaces in the most attractive area

By mapping the workplaces in the concerned area, the right-hand image shows an overrepresentation of the "Commercial" tags, raising questions about the relevance of using these tags. Indeed, the trade sector accounts for half of informal employment according to [28, page V] and [29, page IX]. However, according to [30, page 6], only 19.3% of informal workers have a workplace, with the rest working from home or without a formal workspace. Therefore, I chose not to consider commercial workplaces in the rest of the study, as most are located near or even within the worker's home.

Regarding industrial workplaces, after several tests, it seemed more appropriate to halve their coefficient (10 instead of 21). The results then appeared more realistic, since in some industrial

zones, multiple tags were associated with a single workplace. Additionally, some had to be filtered manually.

B Attractiveness index

The data used in this part comes from detailed Call Detail Records (CDR) collected in 2013 during the Orange Data for Development Challenge [31]. These data cover the activities of more than 9 million anonymized users over a full year. They make it possible to reconstruct individual trajectories and to identify frequent places (home, work) based on the location of cell towers. We used Origin-Destination (OD) matrices derived from the CDR. These matrices, built from trips between 457 zones defined by a Voronoi tessellation centered on cell towers, were extracted using a rigorous methodology [32]:

- Select users with regular activity (at least 4 days of activity during daytime and evening)
- Filter users active for at least 12 one-hour periods during daytime and evening
- Keep only users whose fraction of time spent at home and at work is higher than a threshold of $\delta_h = \frac{1}{3}$

The attractiveness of each area was calculated using the formula below (Formula 1, with F_i^{in} representing the incoming flow and F_i^{out} the outgoing flow), while ignoring the diagonal of the matrix which by definition represents intrazonal movements. The latter represent up to 62.46 % of the total flows. In the article [13], the authors state that up to 80 % of residential and work areas are combined.

$$A_i = \frac{F_i^{\text{in}} - F_i^{\text{out}}}{F_i^{\text{in}} + F_i^{\text{out}}} \quad (1)$$

The results obtained in Figure[15] highlight a concentration of attractiveness in the western areas of Dakar, particularly in the districts of Plateau, Hann-Bel Air, and Fann, which are home to the main economic and administrative activities. In contrast, areas located in the east (such as Pikine or Médina Gounass) present a negative net balance, reflecting a more repulsive dynamic.

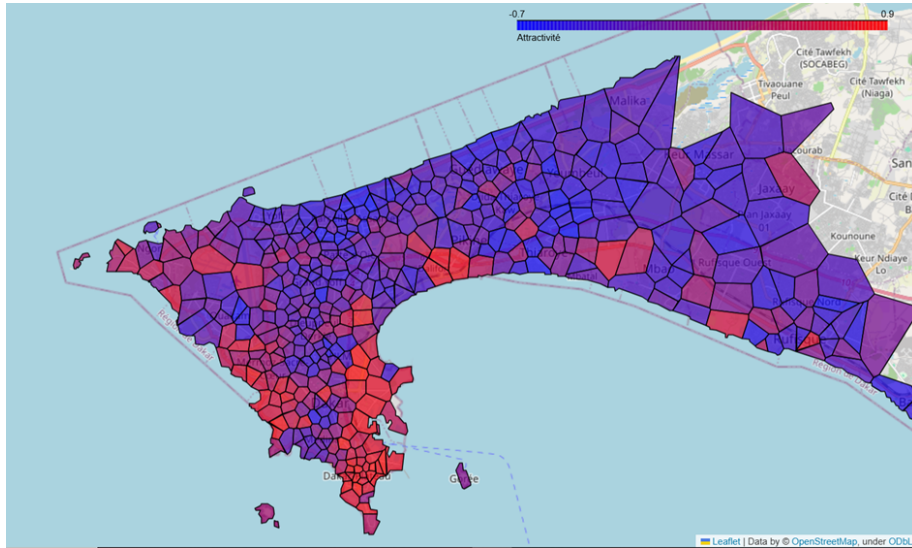
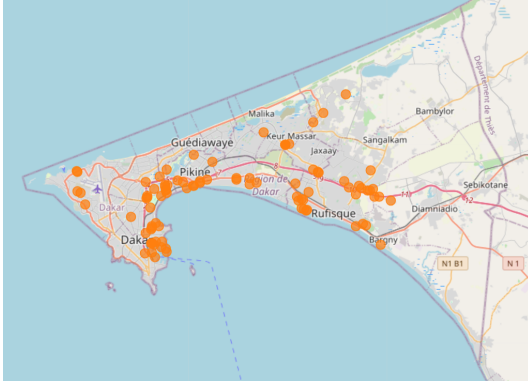
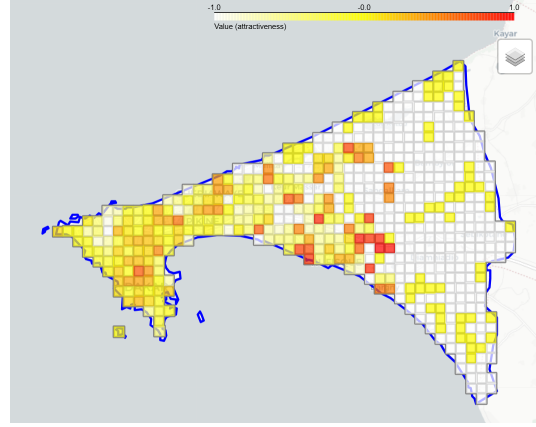


Figure 15: Attractiveness measured from Origin-Destination matrices obtained from CDR datas

The attractiveness index was also calculated from the flows simulated by the EV-PV model using the same metric as in the previous sections, with an implementation directly in the *MobilitySimulator* class. Figure 16b presents the results obtained with uniform vehicle allocation. A clear difference can be observed with the two previous maps. First, the study area is broader here: the study covers the entire Dakar region. Furthermore, in this section, the EVPV model only considers travel by private car. We see a significant impact on industrial zones, particularly towards the east, which I illustrate in Figure[16a] below. Since these are areas where very few cars are allocated due to their low population density, even the smallest workplace is enough to increase the attractiveness index. On the other hand, the geographic areas towards the West (towards Plateau) have negative attractiveness indices despite the high number of workplaces. Looking at the other generated maps, we can see that the outgoing flow is very high towards this area, probably due to the high population density. On the other hand, the incoming flow is a little lower, but it remains in the average values of the nearby areas.



(a) Map of workplaces tagged as "Industrial"



(b) Uniform allocation

Figure 16: Measurement of the attractiveness index with the EV-PV model

C Vehicle distribution by department

The attractiveness mentioned in Appendix [B] is directly influenced by car ownership in the Dakar region, in addition to the population density already considered so far. In 2015, according to [33], vehicle ownership was concentrated among the richest 20% of the population, who owned 57% of vehicles, a slight increase from 53% in 2000. In the Dakar department, 22.4% of households own a car, compared to only 6.4% in Pikine. Central Dakar is particularly notable, housing 73% of cars in 2015, compared to 64% in 2000, also according to [33]. In general, wealth decreases as one moves east, with the exception of Rufisque, located in the far east. Rufisque has seen an increase in car ownership partly due to a new toll road, but mainly because wealthier individuals have moved there to buy homes rather than rent.

Recent ANSD data [12] confirm this geographic disparity. In 2019, the Dakar region's vehicle fleet was distributed as follows: the Dakar department accounted for 76.2% of vehicles (198,748 cars), followed by Pikine (13%, 17,587 cars), Guédiawaye (6%, 11,048 cars), and Rufisque (4%, 3,3498 cars).

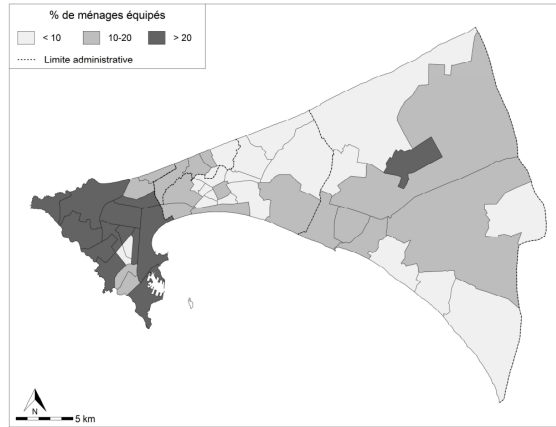


Figure 17: Percentage of households owning private vehicles [33, page 9]

To model these statistics, I extracted the administrative boundaries from the **GADM** files at the departmental level (**level 2**) to assign a corresponding weight to each departmental area in the EVPV model:

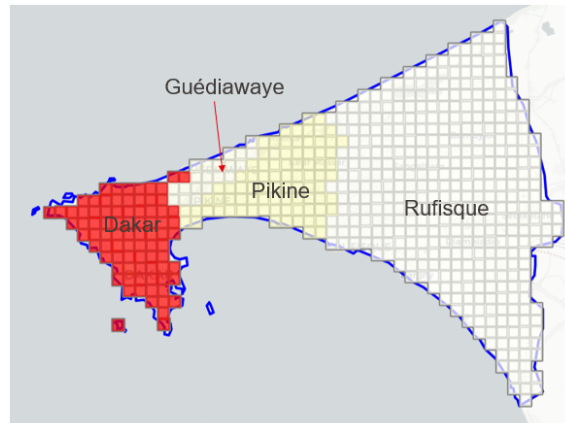


Figure 18: Distribution of private cars by department ;
Dakar : 76%, Pikine : 12%, Guédiawaye : 6%, Rufisque : 4%

D Self-sufficiency and self-consumption and PV capacities

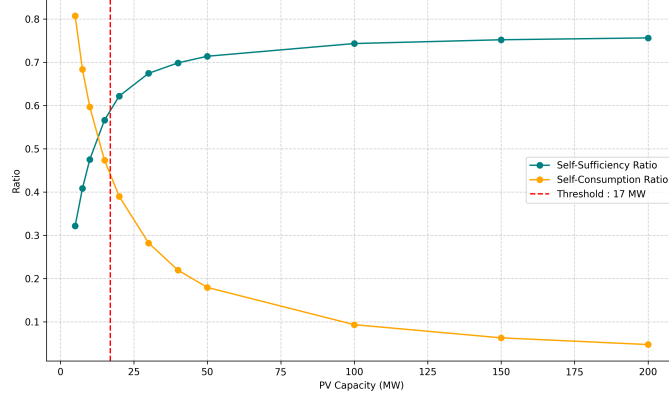


Figure 19: SS and SC ratios for different PV capacities

Self-consumption refers to the proportion of PV energy that is directly consumed by the EV demand, relative to the total PV production. It is calculated as the ratio of the coincident power to the total PV production.

$$\text{Self-Consumption (\%)} = \left(\frac{\int_0^{24} \min(P_{PV}(t), P_{EV}(t)) dt}{P_{PV, \text{ total}}} \right) \times 100 \quad (2)$$

where: $P_{PV}(t)$: PV power at time t , $P_{EV}(t)$: EV charging demand at time t and $P_{PV, \text{ total}}$: total PV production over the day

Self-sufficiency indicates the extent to which the EV energy demand is covered by the PV production. It is calculated as the ratio of the coincident power to the total EV demand.

$$\text{Self-Sufficiency (\%)} = \left(\frac{\int_0^{24} \min(P_{PV}(t), P_{EV}(t)) dt}{P_{EV, \text{ total}}} \right) \times 100 \quad (3)$$

where: $P_{EV, \text{ total}}$: total EV demand over the day

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- [illegible]

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